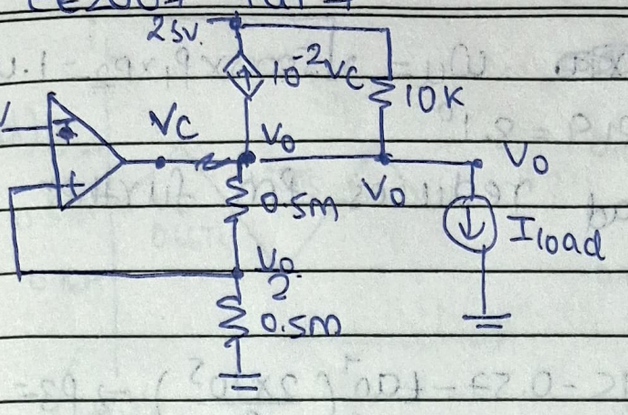
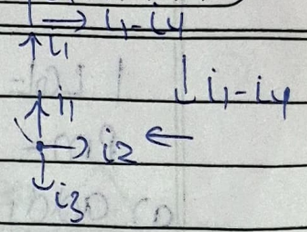
EE2007 tut 7





$$\frac{V_o}{10^6} + 10^{-2}V_C + \frac{V_o - 2.5}{10^4} + I_{load} = 0$$

$10^{-2}V_C = 2.5$ ~~$10^{-2}V_C = 2.5$~~ $10^{-2}V_C = \frac{V_o - 2.5}{10^4}$

$\frac{V_o}{2} = 1.2V \Rightarrow \boxed{V_o = 2.4V}$

$V_C = 0.00026 - 0.1\, I_{load}$

~~$\Rightarrow$~~ $I_{load} = 1mA \Rightarrow \boxed{V_C = -99.24mV}$

$I_{load} = 10mA \Rightarrow \boxed{V_C = -999.24mV}$

$V_C = 100\left(\frac{V_o}{2} - 1.2\right) \Rightarrow V_{out} = \frac{1.20025}{0.500101 + 1/R_{load}}$

$V_{error} = 2.4 - V_{out}$

$R_{load} = \infty \Rightarrow V_{error} = 0.00047V$

② $V_o\, SC + \frac{V_o}{10^6} + \frac{V_o}{10^4} + V_C \cdot g_m = 0$

$\Rightarrow V_o\left(SC + 1/10^6 + 1/10^4\right) = -g_m A(s) V_{test}$

$LG = \frac{V_o/2}{V_{test}} = \frac{-1}{2} \frac{g_m A(s)}{SC + 1/10^6 + 1/10^4}$

$= \frac{-1}{2} \frac{1}{(1 + 5\times10^{-6}s)(1 + 5\times10^{-8}s)(SC + 101\times10^{-6})}$

$= \frac{5\times10^3}{(1 + s/2\times10^5)(1 + s/2\times10^7)(1 + s/p_3)}$

$p_3 = \frac{101\times10^{-6}}{C}$

$|LG|=1 \Rightarrow$ ~~25000~~ $\omega_u = \sqrt{5000 \times p_1 \times p_2} = 1.414 \times 10^5$

$PM = 180 - 90 - 81.9 = 8.1^\circ$

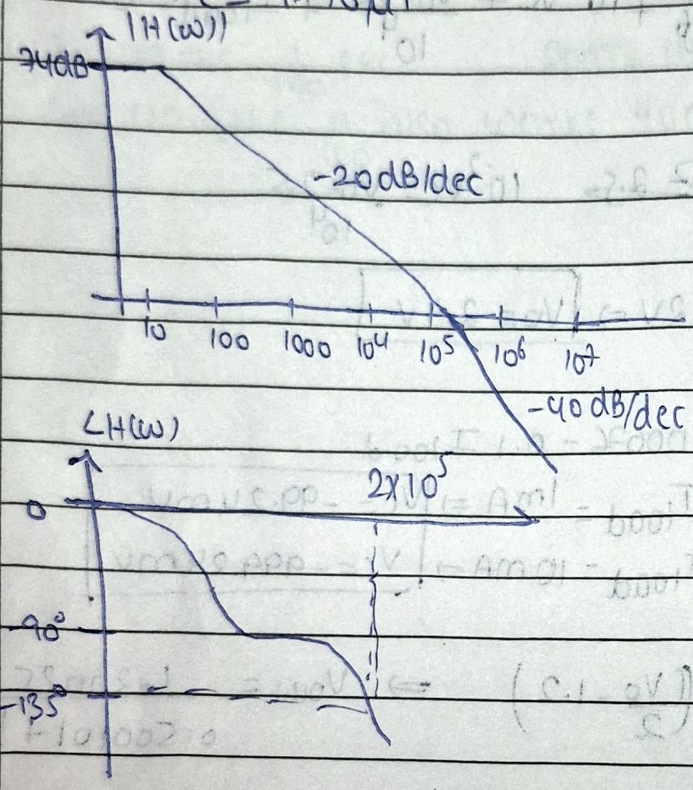
a) adding C_{load} reduces PM further

$\Rightarrow$ ONE

b) $\omega_c = 2 \times 10^5$

$\Rightarrow 45 = 180 - 45 - 0.57 - \tan^{-1}\left(\frac{2\times10^5}{p_3}\right) \Rightarrow p_3 = 56.57$ rad/s

$\Rightarrow C = 1.76\,\mu F$



c) $PM = 0$, $|LG| = 1$

$\Rightarrow C_{LOAD} = 0.1\,pF \,/\, 24.2\,nF$

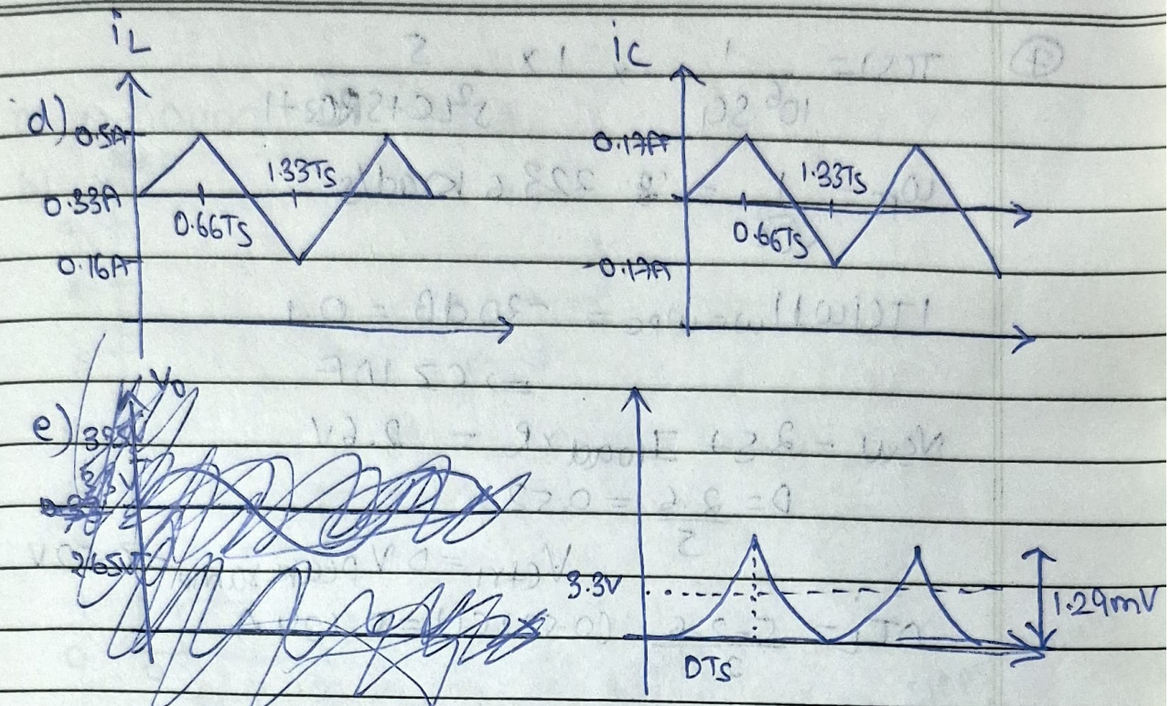
(5) a) $D = \frac{V_{out}}{V_{bat}} = \frac{2.3}{5} = 0.66$

b) $H(s) = \frac{1}{s^2 LC + s(L/R) + 1}$

c) $\Delta i_L = \frac{V_{bat} - V_{out}}{L} \times (DXT) = 0.34A$

$$\Delta V_{out} = \frac{\Delta i_L}{8 f_s C} = 1.29\,mV$$

PageWork
D D M M Y Y Y Y

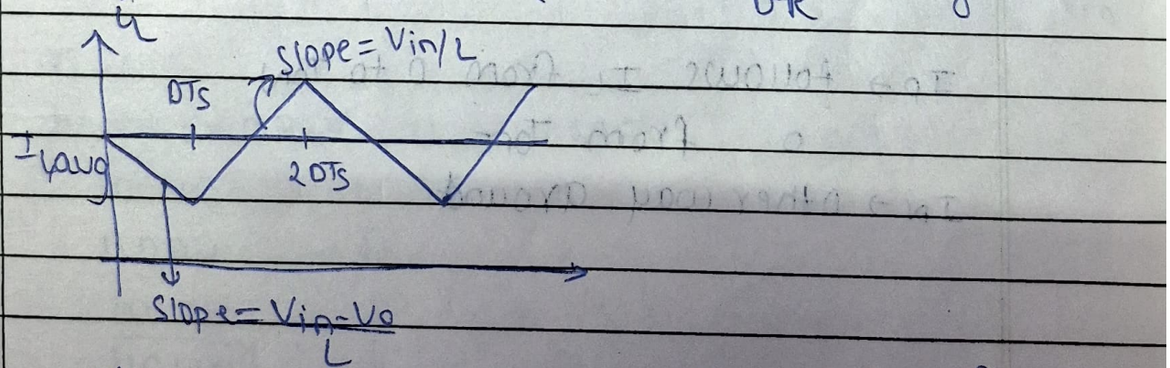


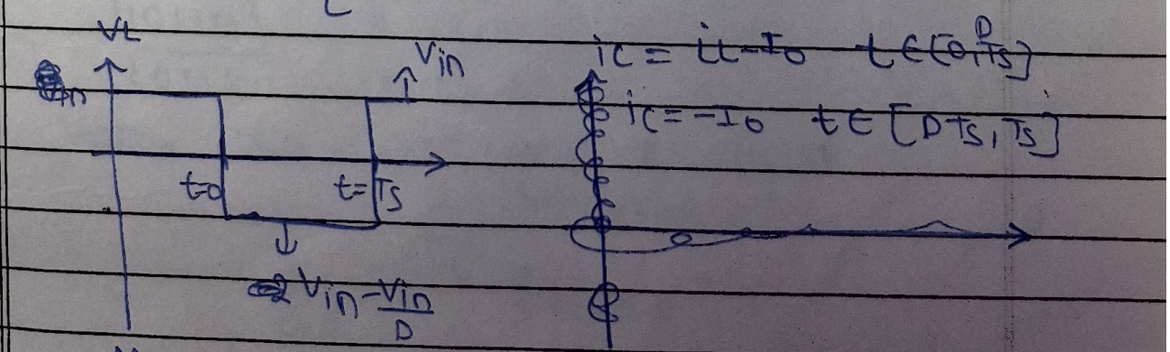
(6) S_1 ON $\Rightarrow V_L = V_{in} - V_o$

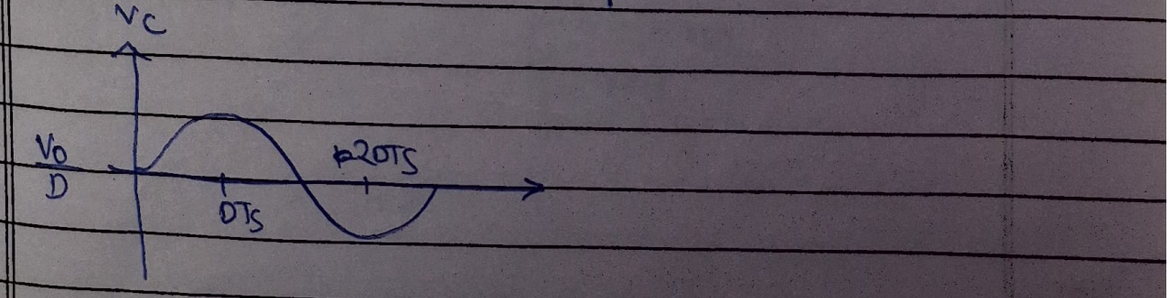
S_2 ON $\Rightarrow V_L = V_{in}$

$$\int_0^{T_S} V_L\,dt = 0 \Rightarrow V_o = \frac{V_{in}}{D}$$

$$P_i = P_{out} \Rightarrow V_{in} I_L = \frac{V_o^2}{R} \Rightarrow I_L = \frac{V_{in}}{D^2 R} \quad (avg)$$







PageWork
D D M M Y Y Y Y

⑦ $T(s) = \frac{1}{10^6 \cdot sC_1} \times 1 \times \frac{5}{s^2LC + sRC + 1}$

$\omega_{pc} = \frac{1}{\sqrt{LC}} = 223.6 \text{ krad/s}$

$|T(j\omega)|_{\omega=\omega_{pc}} = -20\text{dB} = 0.1$

$\Rightarrow C_2 = 1\text{nF}$

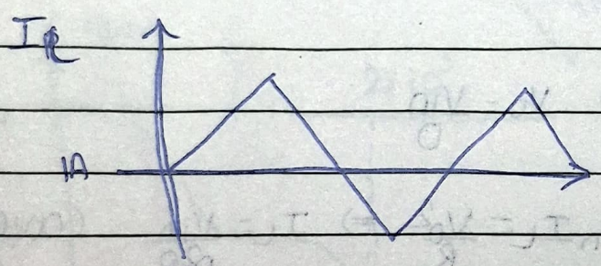
$V_{sw} = 2.5 + I_{load} \times R = 2.6\text{V}$

$D = \frac{2.6}{5} = 0.52$

$V_{ctrl} = D\,V_{peak\text{-}ramp} = 0.52\text{V}$

$\Delta I_L = \frac{5 - 2.6}{2}(0.52)(1) = 0.624\text{ A}$

$\Delta V_o = \frac{\Delta I_L}{8 C f_{sw}} = 0.0078\text{V}$



$I_P \rightarrow$ follows I_L from 0 to T_{ON}

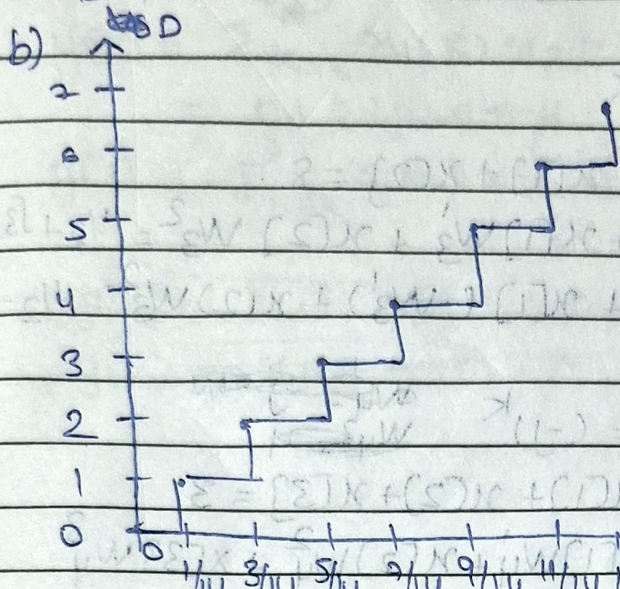
0 from T_{OFF}

$I_N \rightarrow$ other way around

EE2007 tut 8

① a) D = 0000111 D = 3

b)



ideally steps should be 1/8

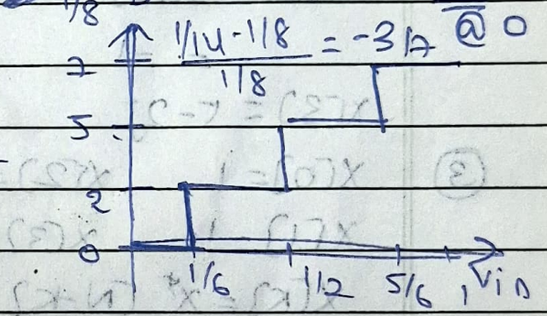
c) LSB = Vref/7 ⇒ DNL = 2/14 − 1/8 = 1/8 − 1/8 = 0

d) V(T1) = V(T2) = 5V/6

V(T3) = V(T4) = T(T5) = V/2

V(T6) = V(T7) = V/6

e) 0000011 D = 2



1/14 − 1/8 = −3/7 @ 0

② a) 101

b) Vth2 = 0.76V Vth3 = 0.625V

③

10000	1
11000	0
10100	1
10110	1
10111	0

Voff = −31.25 μV ⇒ B = 10110

↳ V. small

